CCC-C-441C
September 29, 1969
SUPERSEDING
Int. Fed. Spec. CCC-C-00441B(Army-GL)
April 15, 1969 and
Fed. Spec. CCC-C-441A
June 26, 1964

FEDERAL SPECIFICATION

CLOTH, CORDUROY, COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cotton, corduroy cloth.

1.2 Classification.

1.2.1 Types and classes. This specification covers cotton, corduroy cloth of the following types and classes, as specified (see 6.2):

Type I - Natural.

Class 1 - 10.5 ounce.

Type II - Dyed.

Class 1 - 8.0 ounce.
Class 2 - 7 2 ounce

2 APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics).

Federal Standards:

Fed. Std. No. 4 - Glossary of Fabric Imperfections
Fed. Test Method Std. No 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

2.2 Other publication. The following document forms a part of this specification to the extent specified herein Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D C. 20580)

National Classification Board·

National Motor Freight Classification.

(Application for copies shall be addressed to the National Classification Board, 1616 P Street, N.W., Washington, D.C 20036.)

Uniform Classification Committee·

Uniform Freight Classification

(Application for copies shall be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Ill. 60606.)

3 REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Yarn. The yarn shall be made from cotton, which has been cleaned, carded, drawn and spun into singles for both the warp and filling.

3.3 Color.

3.3 1 Type I. The color of type I cloth shall be natural.

3.3.2 Type II. The color of type II cloth shall be as specified and shall match the standard sample (see 6 3). The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4 4.

3.3 2.1 Matching. The color shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3 2.2 Colorfastness. The dyed and finished cloth shall show colorfastness to combined laundering and bleaching, perspiration, crocking and light equal to or better than the standard sample when tested as specified in 4.4. When no standard sample is established, or when the standard sample is not referenced for colorfastness, the finished cloth shall show "good" colorfastness to combined laundering and bleaching, perspiration, crocking, and light when tested as specified in 4.4.

3.4 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.

TABLE I. Physical requirements

	Weight per square yard (min.)	Yarns per inch minimum		Breaking strength minimum		Wales per inch (min.)
	Ounces	Warp 1/	Filling	Warp	Filling	
Type I						
Class 1	10.5	29	136	86	54	10
2						

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

TABLE I. Physical requirements (con.)

	Weight per square yard (min.)	Yarns per inch minimum		Breaking strength minimum		Wales per inch (min.)
	Ounce	Warp 1/	Filling	Warp	Filling	
Type II						
Class 1	8.0	40	126	60	45	14
Class 2	7.2	46	126	60	45	15

1/ For double tie type, the minimum number of warp yarns shall be twice that specified above (2 ends weaving as 1).

3.4.1 Width. The width shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvage.

3.4.2 Weave. The weave shall be a "W" weave of a single or double tie type.

3.5 Dimensional stability (type II). The finished type II cloth shall not shrink more than 5 percent in the direction of the warp nor more than 3 percent in the direction of the filling when tested as specified in 4.4.

3.6 Nonfibrous materials.

3.6.1 Type I. The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed 13.0 percent when tested as specified in 4.4.

3.6.2 Type II. The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed 2.0 percent when tested as specified in 4.4.

3.7 pH. The pH value of the water extract of the finished cloth shall be no less than 5.5 nor more than 8.5 when tested as specified in 4.4.

3.8 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up in a full width roll as specified in PPP-P-1134.

3.9 Fiber identification. Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.10 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification. The demerit points per 100 square yards shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

4.2.1 Component and material inspection. In accordance with 4 1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item The examination of the end item shall be in accordance with 4.2.2 1 through 4.2.2.3.

4 2 2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in Section I of Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1, except that only those slubs and knots which exceed the maximum limits shown in figure 1 of Fed. Std. No 4 shall be scored No linear yard (increment of one yard on the measuring device of the inspection machine) from any one roll within the sample, shall be penalized more than 4 points. The sample unit for this examination shall be one linear yard. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 35 points The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 52 points. If one roll exceeds 52 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 52 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample X 3600}}{\text{Contracted width of cloth (inches) X total yards inspected}} = \text{Points per 100 square yards}$$

4 2.2 1.1 Demerit points. The demerit points shall be assigned as follows·

For defects up to 3 inches in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur·

Corduroy wales - missing, not properly formed or unevenly spaced.
Baggy, ridgy or wavy cloth.
Uneven pile.
Objectionable odor.
Bald spots throughout
Width less than specified.
Poor dye penetration, mottled, streaky or cloudy.
Excessive neppiness.

4.2.2 2 Examination for length.

4 2.2 2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4 2.2.3 Examination for color and shade. During the yard-by-yard examination, each roll in the sample of type II cloth shall be examined for color and shade Any roll in the sample off shade, shaded side to side, side to center or end to end, shall be cause for rejection of the entire lot represented by the sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

4.2.2.4 Examination for compliance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample are not labeled, ticketed or invoiced in accordance with the Textile Fiber Products Identification Act.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking requirements of section 5 are complied with.

4.4 Testing of the end item. The methods of testing specified in Fed. Std No. 191 wherever applicable and as listed in table II shall be followed The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 1 yard, full width of type I cloth and 2 continuous yards full width of type II cloth. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified The lot size shall be expressed in units of 1 yard. All test reports shall contain the individual values utilized in expressing the final results. The sample size (number of sample units) shall be in accordance with the following:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. Test methods

Characteristic	Requirement paragraph	Test method
Material (cotton yarn)	3.2.1	1200 1/
Labile sulfur	3.3.2	2020
Colorfastness to:		
Light	3.3.2.2	5660
Combined laundering and bleaching	3.3.2.2	5605
Perspiration	3.3.2.2	5680
Crocking	3.3.2.2	5651
Weight	3.4	5041
Wales per inch	3.4	Visual 2/
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Weave	3.4.2	1/ 3/
Dimensional stability.		
Warp	3.5	5550
Filling	3.5	5550

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

TABLE II. Test methods (con.)

Characteristic	Requirement paragraph	Test method
Nonfibrous material		
Type I	3.6.1	2611
Type II	3.6.1	2611
pH	3.7	2811

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ The wales per inch shall be the average of five determinations, rounded off to the nearest whole number

3/ One determination for each sample unit and the result reported as "pass" or "fail".

5. PREPARATION FOR DELIVERY

5.1 Packaging. The packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. The packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in the manufacture of clothing items.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number and date of this specification.
(b) Type and class required (see 1.2).
(c) Color of cloth required (see 3.3.2).
(d) Width of cloth required (see 3.4.1).
(e) Minimum length, if other than specified (see 3.8).
(f) Selection of applicable levels of packaging and packing (see 5 1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 82

Review activities:

Army - MD
Navy - SA
Air Force - 82

Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

GSA-FSS
HEW
COM-NBS
VA
JUS-FPI

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents See section 2 of this specification to obtain extra copies and other documents referenced herein Price 15 cents each

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01760

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRES-EQS
Natick, Massachusetts 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Cloth, Corduroy, Cotton CCC-C-441C

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.